

The Ephemeris

September 2017

Volume 28 Number 03 - The Official Publication of the San Jose Astronomical Association



Sept - Dec 2017 Events

Board & General Meetings

Saturday 9/9, 10/7, 11/4, 12/2

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 9/10, 10/1, 11/5, 12/3

Solar Observing (locations differ)

Sunday 9/10, 10/1, 11/5, 12/3

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 9/1, 9/29, 10/27, 11/24

Astronomy 101 Class

Houge Park 3Q In-Town Star Party

Friday 9/15, 10/13, 11/10, 12/8

RCDO Starry Nights Star Party

Saturday 9/16, 10/14, 12/9

Imaging SIG Mtg

Tuesday 9/19, 10/17, 11/21, 12/19

Astro Imaging Clinics

(at Houge Park)

Saturday 9/23, 12/16

Coder's Meeting

Sunday 9/17, 10/15, 11/19

Quick STArT (by appointment)

Friday 10/6, 12/9

Binocular Star Gazing

No Fall Events

Unless noted above, please refer to the SJAA Web page for specific event times, locations and possible cancelation due to weather.

INSIDE THIS ISSUE

Observing Reports.....	2-3
School Program Post....	4
Pinnacles East.....	5
Cassini	6-7
Astro images.....	8-10
Kid Spot	11
Board of Directors	12
Club Programs	13
Membership Form.....	14

Cover Photo: Crescent Nebula (NGC 6888)

By SJAA Member: Swaroop Shere

SJAA Member Observation Posts

Eclipse Viewing at Houge Park

From Dave Ittner

SJAA hosted an eclipse viewing session this past Monday (August 21, 2017) at Houge Park.

It was publicized on our Meetup group:

www.meetup.com/SJ-Astronomy/events/238378473/

Clicking on that link will bring you to the meetup event posting and allow you to click on the photo symbol to open up the photo album. A number of folks have uploaded some really good pictures.

The event was a huge success. A big thanks to Glenn Newell for assisting me. He came up with the idea of setting up a tent with projection screens inside and displaying a live video feed of the eclipse as well as a live feed of the Totality from NASA (or other sources).

My estimates for the turnout is around 500 folks. We had over 10 scopes set up on Telescope Row (the sidewalk). There were many SJAA club members who brought out their scope with a solar filter. A big heartfelt thanks to those folks. We also had a number of scopes set up impromptu by the public. And we had a number of SJAA members that showed up just to lend a hand (no scopes) Vinod V., Jai P., Arvind H., Hamid S., Swami N.

All in attendance were relieved when the clouds cleared around



9:30 or so. The eclipse began just after 9 and folks could see it when the clouds thinned out. All of us with scopes struggled to get aligned. Apparently clouds + solar filters = 100% blockage of all light.

Photo credit: Swami Nigam

Pictured: Glenn Newell in booth with live video feed of the eclipse.



Photo credit & below comment: Emil

"I don't know about weather but the Sun was so cooperative on the day of the event. I don't remember so many spots for the last two years."

It was an amazing experience to see this eclipse, even if it was only a partial. When it reached maximum, it looked

more like 90% to me than the official 76%.

I can't begin to express my deep gratitude for all who assisted in making this event the success it was.

It is an honor to be a member of SJAA.

Photo Credit: Glenn Newell

A 500+ crowd of eclipse viewers showed up for the event!



Photo Credit: Swami Nigam

Dave Ittner (white shirt) with solar filtered telescope.

SJAA Member Observation Posts

Mendoza Ranch Observing

July 22, 2017

Marilyn Perry (response to Natti and Bob opening up Mendoza Ranch on July 22, 2017 and Chris Kelly response)

Marilyn (July 24th): Thanks, Natti and Bob, for opening up Mendoza Ranch last Saturday evening. I was working on two Astronomical League programs, and I successfully tracked down some faint fuzzies and faint stars making up asterisms. Finding and seeing them was satisfying, but the prettiest views were 1) the view of Saturn through your scope showing how the shadow of the globe on the rings has changed now that it has been a while since opposition and 2) seeing the most nebulosity around the Swan nebula

Christopher Kelly: Jul 26, 2017

Hi Marilyn,
Glad to hear you are working on AL programs. Interested to find out what scope you are using, and what programs!

I have an on again/off again effort to do the H400....I picked up all the Messiers so many years ago, but also working on the Caldwell list.

Marilyn Perry: Jul 27, 2017

Hi Chris,
Good luck to you working on H400 and Caldwell lists. They are very tough lists.

I bought my scope for its weight, 10 lbs. It is a 4" APO Stellarvue refractor, so too small for H400 and Caldwell. But I am an AL program addict. I want to get every pin possible for my equipment. I like to always be searching for something new when I go out observing. I am currently working on Messier and asterisms. I think I could see the final dim Messier galaxies with an autumn trip and a spring trip to Pinnacles.

The asterism program took some research to figure out what I was supposed to be looking for. My current recent favorite is "Bent Fan", aka "red necked emu". It is in Cygnus, its brightest star is

b3 Cyg, the asterism is very bright, and whereas I used to get lost sailing down Cygnus from Deneb to Albireo, now I have a reference point so I know where I am. I found something very cute last night, tennis ball and racket in Aquila where the tennis ball is HD 187570, brightest star in asterism. The racket is very easy to pick out and easy to see in town.

Previous AL programs I have completed are binocular double star, binocular Messier, double star, carbon star, Lunar, Lunar II, Urban, Galileo, and Solar system. I had to borrow club's 8" dob to see a couple of galaxies in the Urban program.

There were times I thought I would not be able to complete Lunar II because some features were too small, but I kept trying month after month, and when shadows were just right, I finally saw all features but Beer cantena. Lunar II has some optional activities, so you don't need to see everything.

I started working on Solar System just as I was finishing Galileo, and a couple of the Jupiter activities were very similar. It was very interesting to me to see how different the Jupiter moons looked a couple of years apart depending on how the orbits of the moons were lined up with respect to my view.



SJAA School Star Party Central

From Tom Piller

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

The school star party coordinator for SJAA is Jim Van Nuland. Jim has been involved with the school program for over 3 decades. Gary Mitchell, Paul Mancuso, Terry Kahl, Marilyn Perry, Gordon, Teruo Utsumi and a few more have also been with the program for many years and / or volunteer on a regular basis.

In support of the program, SJAA volunteers provide telescopes, create observing lists tailored to what's up that time of year, provide instruction on viewing the night sky, work with each school to make decisions on who will attend. And then there is the coordination on the viewing location, lights, sprinklers and always the weather.

The program pretty much operates during the 10 months of the school year with the most active months occurring from November through April. 2016-2017 metrics produced 39 scheduled school star parties; and out of that 21 events which were successful and 18 which were cancelled due to weather. That's not bad considering all of the rain we had in the Bay Area last winter. I have heard Jim announce at club meetings that there might be as many as 3 star parties in one week. That's a lot of star parties!

Many thanks to Jim and the faithful SJAA volunteers who help out to provide this valuable and successful community outreach program.

School star parties are not open to the public. Though school grounds are open during these star parties, it is the school, not SJAA that is the host.



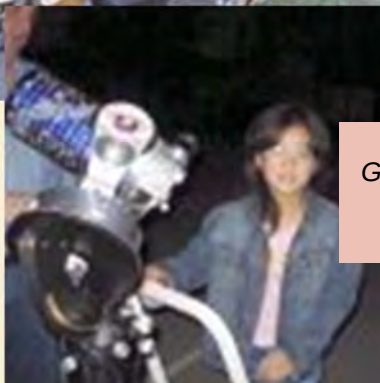
Photo (left): Jim Van Nuland setting up his 8" Newtonian scope which he purchased in 1982. Since then he has added the 4.25 inch f/4 scope, an adjustable north leg, an inverter to run the 120v motor from battery, and he can vary the running speed and make small declination moves with a joystick. Per Jim; optically excellent!



Jim's 8-inch, children waiting patiently



When parents wander away!



Gordon's Questar and student



Pinnacles East Star Party

From Tom Piller

Under the leadership of Ed Wong, on July 22, 2017, SJAA hosted it's 1st Pinnacles East Public star party in conjunction with the National Park Service (NPS).

SJAA had 10 astronomers, the skies were dark and approximately 200 park visitors attended the event. The Park Rangers gave an introductory talk about Pinnacles and how excited they were to have SJAA hosting the event with their telescopes. Ed Wong followed up with a short talk about the night sky. The star party was considered a big success and two more events are being planned for Fall 2017.

The horizons are a little on the high side but there were no observable light domes and only a little bit of ambient light coming from some park structures nearby. The crowds were treated to views of Jupiter, Saturn, rich views of the Milky Way, tons of globulars, nebulae, star clusters, the usual galaxies, and lots of double stars; it was an awesome night.

One of the nice things about the Pinnacles East site [behind the General Store] is that the parking lot could easily handle around 15-20 scopes and the public is required to park either at their campsites or in the General Store parking lot which helps keep the white light and traffic confusion to a minimum.

Ed has also been working with the NPS Interpretive Ranger to enable Pinnacles East to become a certified Dark Sky Site by the International Dark Sky Association. To this end, SJAA has made several exploratory observing trips to Pinnacles East, taking SQM light readings to document the night sky.

Pinnacles East is approximately a 2 hour drive south of the San Jose area, or 1.5 hour drive from the Monterey Bay area. Some of the astronomers planned ahead and obtained a campsite while most packed up and drove home after the observing session; so lots of options.

NASA Announces Cassini End-of-Mission Media Activities



NASA's Cassini spacecraft is shown during its Sept. 15, 2017, plunge into Saturn's atmosphere in this artist's depiction. Cassini will use its thrusters to keep its antenna pointed at Earth for as long as possible while sending back unique data about Saturn's atmosphere.

Credit: NASA/JPL-Caltech

On Sept. 15, NASA's Cassini spacecraft will complete its remarkable story of exploration with an intentional plunge into Saturn's atmosphere, ending its mission after nearly 20 years in space. News briefings, photo opportunities and other media events will be held at NASA's Jet Propulsion Laboratory (JPL) in Pasadena, California, and will air live on NASA Television and the agency's website.

Launched in 1997, Cassini arrived in orbit around Saturn in 2004 on a mission to study the giant planet, its rings, moons and magnetosphere. In April of this year, Cassini began the final phase of its mission, called its Grand Finale -- a daring series of 22 weekly dives between the planet and its rings. On Sept. 15, Cassini will plunge into Saturn, sending new and unique science about the planet's upper atmosphere to the very end. After losing contact with Earth, the spacecraft will burn up like a meteor. This is the first time a spacecraft has explored this unique region of Saturn -- a dramatic conclusion to a mission that has revealed so much about the ringed planet.

Cassini flight controllers will monitor the spacecraft's final transmissions from JPL Mission Control. Interviews with mission engineers and scientists will be available for media.

Cassini Media Events and Schedule

(All media teleconferences and NASA TV news conferences will be available on the agency's website, and times are subject to change)

Wednesday, Sept. 13

1 p.m. EDT -- News conference from JPL with a detailed preview of final mission activities (also available on NASA TV and online)

Thursday, Sept. 14

10 a.m. to 3 p.m. PDT -- NASA Social -- onsite gathering for 30 pre-selected social media followers (JPL-accredited media may also attend). Events will include a tour, and a speaker program that will be carried on NASA TV and online.

After 2 p.m. PDT -- Media tours of Mission Control

About 8 p.m. PDT -- Final downlink of images expected to begin (streamed online only)

Friday, Sept. 15: End of Mission

7 to 8:30 a.m. EDT -- Live commentary on NASA TV and online. In addition, an uninterrupted, clean feed of cameras from JPL Mission Control, with mission audio only, will be available during the commentary on the NASA TV Media Channel and on Ustream.

About 8 a.m. EDT -- Expected time of last signal and science data from Cassini

9:30 a.m. EDT -- Post-mission news conference at JPL (on NASA TV and online)

For online streaming, visit: <https://www.nasa.gov/live>

To watch the news conferences online, visit: <https://www.nasa.gov/live> <http://www.youtube.com/nasajpl/live>

Saturn-lit Tethys

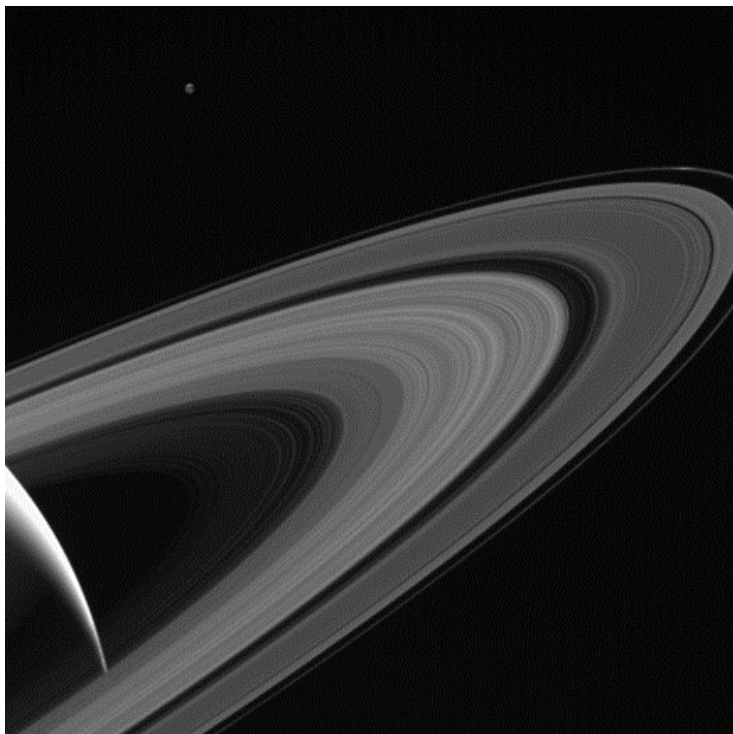
Cassini gazes across the icy rings of Saturn toward the icy moon Tethys, whose night side is illuminated by Saturnshine, or sunlight reflected by the planet.

Tethys was on the far side of Saturn with respect to Cassini here; an observer looking upward from the moon's surface toward Cassini would see Saturn's illuminated disk filling the sky.

Tethys was brightened by a factor of two in this image to increase its visibility. A sliver of the moon's sunlit northern hemisphere is seen at top. A bright wedge of Saturn's sunlit side is seen at lower left.

This view looks toward the sunlit side of the rings from about 10 degrees above the ring plane. The image was taken in visible light with the Cassini spacecraft wide-angle camera on May 13, 2017.

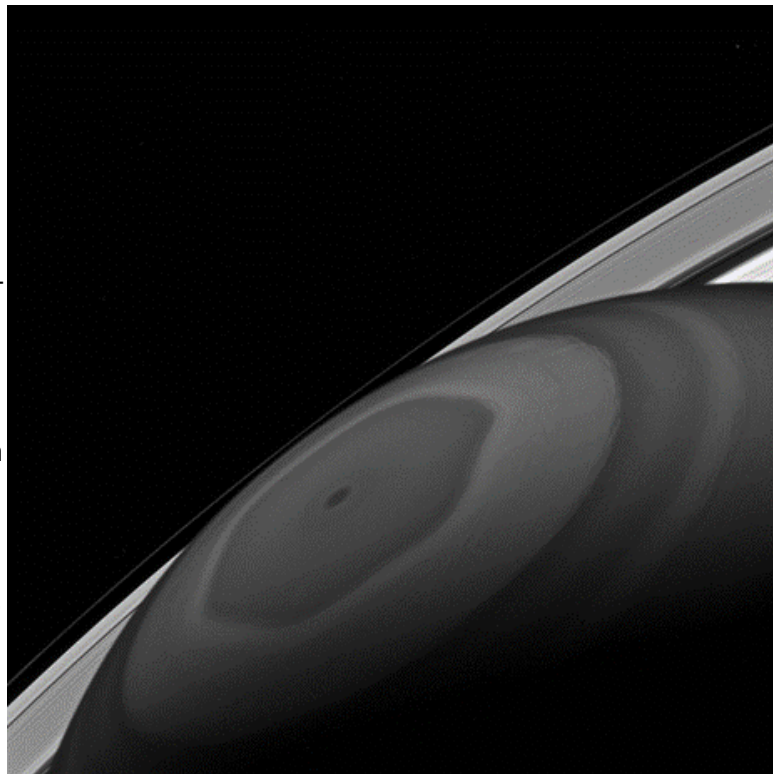
The view was acquired at a distance of approximately 750,000 miles (1.2 million kilometers) from Saturn and at a Sun-Saturn-spacecraft, or phase, angle of 140 degrees. Image scale is 43 miles (70 kilometers) per pixel on Saturn. The distance to Tethys was about 930,000 miles (1.5 million kilometers). The image scale on Tethys is about 56 miles (90 kilometers) per pixel.



The Cassini mission is a cooperative project of NASA, ESA (the European Space Agency) and the Italian Space Agency. The Jet Propulsion Laboratory, a division of the California Institute of Technology in Pasadena, manages the mission for NASA's Science Mission Directorate, Washington. The Cassini orbiter and its two onboard cameras were designed, developed and assembled at JPL. The imaging operations center is based at the Space Science Institute in Boulder, Colorado.

Credit: NASA/JPL-Caltech/Space Science Institute

The Hexagon



The north pole of Saturn sits at the center of its own domain. Around it swirl the clouds, driven by the fast winds of Saturn. Beyond that orbits Saturn's retinue of moons and the countless small particles that form the ring.

Although the poles of Saturn are at the center of all of this motion, not everything travels around them in circles. Some of the jet-stream patterns, such as the hexagon-shaped pattern seen here, have wavy, uneven shapes. The moons as well have orbits that are elliptical, some quite far from circular.

This view looks toward the sunlit side of the rings from about 26 degrees above the ring plane. The image was taken with the Cassini spacecraft wide-angle camera on Dec. 2, 2016 using a spectral filter which preferentially admits wavelengths of near-infrared light centered at 890 nanometers.

The view was acquired at a distance of approximately 619,000 miles (996,000 kilometers) from Saturn. Image scale is 37 miles (60 kilometers) per pixel.

The Cassini mission is a cooperative project of NASA, ESA (the European Space Agency) and the Italian Space Agency. The Jet Propulsion Laboratory, a division of the California Institute of Technology in Pasadena, manages the mission for NASA's Science Mission Directorate, Washington. The Cassini orbiter and its two onboard cameras were designed, developed and assembled at JPL. The imaging operations center is based at the Space Science Institute in Boulder, Colorado.

For more information about the Cassini-Huygens mission visit <http://saturn.jpl.nasa.gov> and <http://www.nasa.gov/cassini>. The Cassini imaging team homepage is at <http://ciclops.org>.

Credit: NASA/JPL-Caltech/Space Science Institute



Orion: Belt, Flame, and Horsehead

Image Credit & Copyright: Rogelio Bernal Andreo (Deep Sky Colors)



Iris Nebula (NGC 7023)
From: Bruce Braunstein

Location: Stellarvue Star Party 2017 (June 21st – June 24th)

Telescope or lens: Astro Tech AT65EDQ 65mm f/6.5 ED quadruplet

Camera: SBIG STT 8300M

Mount: Software Bisque Paramount MyT

Software: Software Bisque TheSkyX Pro, Pixinsight 1.8, Photoshop CS6

Accessory: SBIG filter wheel FW8S-STT

Resolution: 3218x2384

Frames:

Astrodon Blue Tru-Balance E-Series Gen 2: 15x450" -20C bin 2x2

Astrodon Green Tru-Balance E-Series Gen 2: 15x450" -20C bin 2x2

Astrodon Luminance E-Series: 9x900" -20C bin 1x1

Astrodon Red Tru-Balance E-Series Gen 2: 15x450" bin 2x2

Integration: 7.9 hours

Darks: ~15

Flats: ~20

Bias: ~200

M8 The Lagoon Nebula
From: Nikola Nikolov

Location: Astroimaging Workshop at Coyote Valley, CA
Date: July 22, 2017

Telescope: 6" Orion Newton
Mount: Orion Sirius EQ, with autoguiding.
Software: DSS, Pixinsight, and DxO
CCD; extra narrow band H-alpha.
Exposure: 20 minutes



Photo: Solar eclipse
NASA

Kid Spot Jokes:

- ♦ **Why did the sun get straight A's?**
Because it's so bright!
- ♦ **Why is the moon like a dollar ?**
Because it has four quarters.
- ♦ **If a meteorite hits a planet what do we call the ones that miss ?**
Meteorwrongs!

Kid Spot Quiz:

1. **What is the average duration of "totality" in a total solar eclipse?**
2. **What is the phase of the moon during a solar eclipse?**
3. **What is the maximum number of solar eclipses that occur in a calendar year?**

100 Years of Astronomical Discovery

Excerpts From Astronomy Magazine September 2016 Edition

1920 Astronomers directly measure the diameter of the red super giant star Betelgeuse (Alpha Orionis)

1921 A major geomagnetic storm occurs on Earth due to a coronal mass ejection from the Sun from May 13 to 15. Telegraph systems were severely damaged as a result.

1921 Albert Einstein wins the Nobel Prize in physics for his discovery of the photoelectric effect (not for the theory of relativity).

1922 Canadian astronomer John Stanley Plaskett discovers what will become known as Plaskett's Star, a binary system more than 100 times as massive as the Sun.

1924 English astronomer Arthur Eddington proposes the mass luminosity relationship: Stars with more mass are more luminous.

1925 Cecilia Payne-Gaposchkin submits a doctoral thesis hypothesizing that hydrogen and helium are the main constituents of stars.

1926 Goddard launches the first liquid propellant rocket in Auburn Massachusetts.

1928 The IAU formalizes the constellation boundaries.

1929 Hubble's Constant, a value to measure the expansion rate of the universe, is first proposed; 70 km/sec/Mpc (1Mpc = 32.6 million light years). The first person to propose it was George Lemaitre in 1927 but Hubble used observational evidence to confirm the law and by calculating a more accurate value for the relationship between a galaxy's distance and its recessional speed. Astronomers christened this number "Hubble's constant."

1930 Astronauts Neil Armstrong and Buzz Aldrin are born.

Kid Spot Quiz Answers:

1. 2-3 minutes
2. New moon
3. 5

From the Board of Directors

Announcements

- ◇ Fall Swap Meet will be held on Sunday, October 29th. Please see the Web Page for additional information.

Board Meeting Excerpts

May 6, 2017

In attendance

Teruo Utsumi, Bill O'Neil, Rob Chapman, Sukhada Palav, Vini Carter, Glenn Newell, Wolf Witt, Swami Nigam
Excused: Rob Jaworski
Guest: Dave Ittner

Hogue Park Application to San Jose

The application is due May 30th. Bill O'Neil is looking into and will follow up on the requirements.

SJAA Membership Database Coders and IT update

Teruo Utsumi led discussion on a Website hosting change along with a membership database / application. Paypal application is in testing.

AskSJAA mailing list

Teruo Utsumi has created an AskSJAA mailing list to field questions from web page inquiries. Several volunteers stepped up to be on the list.

Cleanup/Repair of 24" scope

Dave Ittner submitted a budget for repairs required before the 24" donated scope can be sold as a complete package. The Board approved the request.

Meetup Announcement Guidelines

Bill O'Neil expressed a need to set some guidelines for how far out into the future Meetup drafts and announcements need to be placed on the Meetup site. It was suggested and agreed that 4 weeks works.

June 10, 2017

In attendance

Teruo Utsumi, Rob Chapman, Sukhada Palav, Vini Carter, Glenn Newell, Wolf Witt, Rob Jaworski

Excused: Bill O'Neil, Swami Nigam
Guest: Muditha Kanchana, Emi, Rashi

Hogue Park Application to San Jose

Per Teruo Utsumi, in order to fully execute the contract for use of Hogue Park, the City is requiring background check and tuberculosis test for any member who might help to host an event at Hogue Park. SJAA discussed pursuing a less burdensome path with the City. Outcome is TBD

Help for SJAA Tasks

Teruo Utsumi requested that Board recruiting for SJAA member help with programs be discussed and announced at every meeting. An example is an ongoing need for more scopes for the school program Jim Van Nuland runs.

New Scope for School Program

Rob Jaworski led discussion on progress in setting requirements and budget for a new telescope / mount for Jim Van Nuland for the School Star Party Program Jim leads. The new budget was approved by all.

SJAA Physical Address Change

Teruo Utsumi brought up the need to change the physical address from the current personal address to one that can be more permanent and still qualifies for government paperwork, etc. Rob Jaworski is checking into some options. This also ties into cleaning up the club's location as it shows on Google Maps.

July 8, 2017

In attendance

Teruo Utsumi, Rob Chapman, Bill O'Neil, Glenn Newell, Vini Carter, Swami Nigam, Sukhada Palav, Rob Jaworski
Absent Excused: Wolf Witt
Guest: Dave Ittner

Bi-Law Amendments

Rob Jaworski motioned to remove Article 12, Section 1, Part C in its entirety. This will remove any mention of a youth or student membership. Further, Section 2 will be amended to state that only individuals 18 and older can be members. These changes being consistent with current operation of the Club. The motion was approved by all.

Website and membership forms will be updated to reflect the changes.

Open Space Authority (OSA) Requirements

Bill O'Neil distributed a complete list of OSA requirements for future events. The RCDO Stary Nights event has become so popular the OSA staff can no longer handle the crowds and some changes need to be made; example...limiting crowd size. Discussion on how to best manage the events is ongoing.

SJAA 2018 calendar

Per Teruo Utsumi it is time to start generating the Club's event list for 2018. Teruo has created a Scheduler's App to handle much of the heavy lifting but program leaders will still need to review the schedule for errors and special cases like holiday conflicts.

August 5, 2017

In attendance

Teruo Utsumi, Rob Chapman, Bill O'Neil, Vini Carter, Swami Nigam, Sukhada Palav, Wolf Witt
Absent Excused: Glenn Newell, Rob Jaworski
Guest: Dave Ittner

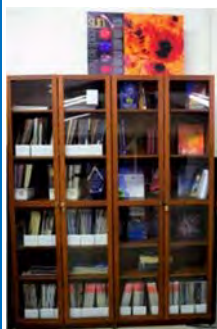
Hogue Park Application to City of San Jose

Teruo Utsumi provided status: contract sent out for Board review and comment, Rob Jaworski requested a recommendation for a CPA to provide support documentation for the non-profit status certification, bring up background check and tuberculosis inoculation with District 9 council member for possible relief (pending).

Open Space Authority (OSA) Requirements

Bill O'Neil chaired discussion on future SJAA involvement in the OSA RCDO Program and how to move forward; with no resolution. Teruo Utsumi will arrange to meet with the OSA representative in charge of the program to voice concerns over SJAA partnering with OSA and being involved in their planning / decision process early on.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly

session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Teruo Utsumi
Vice Pres/Dir:	Bill O'Neil
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Vini Carter
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Sukhada Palav
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Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Sukhada Palav
Loaner Program:	Muditha Kanchana
Memberships:	Anand Rajagopalan
Publicity:	Rob Jaworski
Questions:	
Quick START	Dave Ittner
Solar & Starry Nights:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with online Ephemeris
- ☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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